

Validation Plan - 3-Axis SCARA Robot Embedded System

ESP32-S3 | 3 x NEMA 17 | A4988 Drivers | Limit Switches | 12 V Electromagnet

1. Purpose and System Under Validation

This validation plan defines how the main performance indicators of the SCARA robot will be measured and how project progress will be documented. The system under validation is the architecture defined previously: an ESP32-S3 controls two rotational axes and one vertical Z axis through three A4988 drivers and NEMA 17 stepper motors. Three limit switches provide homing/travel references, and a 12 V electromagnet controlled through a MOSFET is used as the end effector.

2. Selected KPIs and Validation Table

KPI	Measured Variable	Unit	Instrument	Procedure	Expected Value	Acceptance Criterion
Positioning accuracy	Difference between commanded and actual end-effector position after a point-to-point move	mm	Steel ruler/caliper, marked reference grid	Command the robot to 5 predefined XY positions. Measure actual position three times at each point and calculate absolute position error.	≤ 3 mm absolute error	PASS if the maximum measured absolute error at all test points is ≤ 3 mm.
Homing repeatability	Variation of the reference position after repeated homing cycles	mm	Digital caliper or fixed reference mark	Run the homing routine 10 times from different initial positions. Measure the final reference position after each cycle.	$\leq \pm 1$ mm variation	PASS if every homing result remains within ± 1 mm of the reference position.
Command response latency	Time from command transmission to the beginning of robot response	ms	PC serial monitor + firmware timestamp / logic analyzer if available	Send 20 motion or electromagnet commands through USB/UART and record the time between command reception and output activation.	≤ 100 ms	PASS if 95% or more of the commands begin execution within 100 ms.
Power consumption	Input current and electrical power during idle and normal multi-axis operation	A and W	Digital multimeter or DC power meter	Measure current at 12 V while idle and during a representative movement cycle with the electromagnet operating.	Normal operation < 5.0 A; peak < 6.0 A	PASS if continuous current remains below 5.0 A and short peaks do not exceed the 6.0 A supply rating.
Operational stability	Ability to complete repeated cycles without reset, lost steps, false limit events, or uncontrolled motion	Successful cycles / total cycles (%)	Visual inspection, serial log, reference marks	Run 20 complete pick-and-place cycles using the same trajectory and inspect for resets, position drift, or unexpected stops.	100% of 20 cycles without critical fault	PASS if all 20 cycles finish without controller reset, uncontrolled motion, or observable lost-step position error.
Electromagnet pick reliability	Successful pickup, transport, and release of the selected ferromagnetic test piece	%	Test piece + cycle counter	Perform 20 pickup/transport/release attempts using the intended test piece and normal motion profile.	$\geq 95\%$ successful cycles	PASS if at least 19 of 20 attempts are completed without dropping or failing to release the part.

3. Test Cases

TC-01 - Homing and Limit-Switch Test

Objective	Verify that the robot establishes a repeatable reference position and stops safely at each configured limit.
Initial conditions	Robot powered on; axes positioned away from home; limit switches connected; electromagnet OFF.
Procedure	1) Start the homing routine. 2) Observe each axis moving toward its switch. 3) Verify that motion stops when the switch activates. 4) Repeat 10 times from different initial positions. 5) Measure the final home position.
Expected result	All three axes reach home without collision and the final reference position is repeatable.
Pass/Fail criterion	PASS if every cycle completes correctly and the final reference position remains within +/-1 mm.

TC-02 - Point-to-Point Positioning Test

Objective	Evaluate positioning accuracy at representative points in the SCARA workspace.
Initial conditions	Robot homed; no payload or standard test payload installed; reference grid fixed to the work surface.
Procedure	1) Home the robot. 2) Command five predefined XY/Z target positions. 3) Measure actual end-effector position. 4) Repeat each point three times. 5) Record absolute error.
Expected result	The end effector reaches each commanded point with small and repeatable error.
Pass/Fail criterion	PASS if the maximum absolute position error is <= 3 mm at every test point.

TC-03 - Power Consumption Test

Objective	Verify that the 12 V / 6 A power architecture supports normal robot operation without excessive current draw or voltage instability.
Initial conditions	Multimeter or DC power meter connected at the 12 V input; robot homed; representative payload available.
Procedure	1) Measure idle current. 2) Run simultaneous/representative axis motion. 3) Activate the electromagnet during the cycle. 4) Record normal and peak current. 5) Observe for ESP32-S3 resets or driver faults.
Expected result	The system remains stable and current stays within the power-supply capability.
Pass/Fail criterion	PASS if normal current is < 5.0 A, peaks remain < 6.0 A, and no controller reset occurs.

TC-04 - Electromagnet Pick-and-Place Test

Objective	Verify reliable pickup, transport, and release of the selected ferromagnetic part.
Initial conditions	Robot homed; test piece placed at the defined pickup location; electromagnet and MOSFET stage connected.
Procedure	1) Move to pickup point. 2) Activate electromagnet. 3) Lift and transport the part. 4) Move to drop point. 5) Deactivate electromagnet. 6) Repeat 20 times.
Expected result	The robot completes the pick-and-place sequence without dropping the part and releases it at the target.
Pass/Fail criterion	PASS if at least 19 of 20 cycles are successful.

TC-05 - Continuous Stability Test

Objective	Verify stable operation during repeated robot cycles.
Initial conditions	Robot homed; normal firmware loaded; normal payload and power supply connected.
Procedure	1) Run 20 complete automatic cycles. 2) Monitor serial messages and physical position. 3) Record resets, false limit events, lost steps, overheating, or unexpected stops.
Expected result	The robot completes all cycles with consistent position and no critical fault.

Pass/Fail criterion	PASS if all 20 cycles finish without reset, uncontrolled motion, or observable lost-step error.
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4. Technical Logbook - First Entry

Field	Entry
Date	September 19, 2026
Team members	_____
Current project status	System architecture defined; validation planning stage. The mechanical SCARA concept, ESP32-S3 controller, three NEMA 17 motors with A4988 drivers, limit switches, 12 V electromagnet, MOSFET switching stage, and 12 V / 6 A power architecture have been selected.
Work completed	Defined the embedded-system architecture; replaced the original Arduino-based concept with ESP32-S3 control; selected the electromagnet as the end effector; created the hardware/software partition, preliminary BOM, technical-risk analysis, validation KPIs, acceptance criteria, and initial test cases.
Problems found	The previously available 12 V / 4.5 A supply provides limited margin compared with the project requirement. The exact current rating of the final NEMA 17 motors and the holding force/current of the selected electromagnet still need to be confirmed. A4988 current limits must also be adjusted before full-load testing.
Decisions made	Use an ESP32-S3 as the central controller; use three NEMA 17 motors and three A4988 drivers; use a 12 V electromagnet instead of a servo gripper; specify a 12 V / 6 A supply for the final architecture; use limit switches for homing and travel limits; use USB/UART as the initial communication interface.
Next task	Assemble and verify the power/control electronics on the bench, configure A4988 current limits, test the limit switches and electromagnet driver, and then execute TC-01 before motion-accuracy and endurance tests.

5. Project Repository Structure

The project folder is organized as follows:

```
SCARA_Validation_Project/  
|-- hardware/  
|-- firmware/  
|-- documentation/  
|-- tests/  
|-- README.md  
`-- REPOSITORY_LINK.txt
```

The online repository link cannot be generated until the team uploads this prepared folder to the repository platform required by the instructor. A REPOSITORY_LINK.txt file is included so the final shareable URL can be inserted after upload.

6. README Summary

The README included in the project root explains the project purpose, main hardware, repository structure, validation strategy, recommended test order, current status, and an electrical safety note.

7. Analysis Questions

1. Which KPI is the most important for your project and why?

Positioning accuracy is the most important KPI because the main purpose of the SCARA robot is to move an end effector to defined pickup and drop locations. If the robot cannot reach the required positions consistently, the pick-and-place task will fail even if the electronics and communication work correctly.

2. What instrument will be required to measure it?

A digital caliper is preferred for short-distance measurements, together with a fixed reference grid or ruler on the work surface. For a first prototype, these tools provide a realistic and low-cost way to compare commanded and actual positions.

3. What result would indicate that the prototype does not meet the requirements?

The prototype does not meet the preliminary requirement if the absolute positioning error exceeds 3 mm at any validation point, if homing varies by more than ± 1 mm, or if a critical fault such as a controller reset or uncontrolled motion occurs during the stability test.

4. Which test should be performed first?

The homing and limit-switch test should be performed first because a reliable reference position and safe travel limits are required before evaluating positioning, power consumption during motion, or automatic pick-and-place cycles.

5. Why is a technical logbook useful during an engineering project?

A technical logbook provides traceability. It records what was tested, which problems were found, what design decisions were made, and what must be done next. This makes troubleshooting easier and allows the team to justify changes using documented evidence.

8. Validation Sequence

The recommended sequence is: electrical inspection -> homing/limit validation -> point-to-point positioning -> power measurement -> electromagnet pick-and-place -> continuous stability. This order reduces risk because safe reference motion is verified before more demanding automatic tests.

Note: The numerical acceptance criteria in this initial plan are preliminary engineering targets for the course prototype. They should be updated only if later mechanical dimensions, payload requirements, or instructor requirements define stricter values.